MOHIT KUMAR

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SUMMARY

Motivated B.Tech CSE (AI/ML) student with hands-on experience in Python, Machine Learning, and Data Science. Proficient in building end-to-end ML projects including CNN-based computer vision systems and predictive models. Completed 100+ hours of structured training in Python, NumPy, Pandas, and AI/ML workflows. Seeking to apply technical skills in a real-world AI/ML or Data Science role.

EXPERIENCE

Open to entry-level opportunities in Machine Learning / AI / Data Science. Practical experience demonstrated through projects below.

TECHNICAL SKILLS

Languages: Python
Data Analysis & Visualization: Pandas, NumPy, Matplotlib
Machine Learning: Scikit-learn, Supervised Learning, Unsupervised Learning, Regression, Classification
Deep Learning: PyTorch, TensorFlow, CNN
Computer Vision: OpenCV, Object Detection, Image Classification & Preprocessing
Generative AI: LangChain, RAG & Embeddings, LLMs, Agentic AI
APIs & Frameworks: FastAPI
Tools & Platforms: Git, GitHub, Google Colab
Soft Skills: Communication, Problem-Solving, Teamwork, Adaptability

PROJECTS

RAG Evaluation Suite | LangChain, ChromaDB, DeepEval, OpenRouter, LLM APIs

- Built an end-to-end RAG pipeline (retriever + cross-encoder reranker + LLM generator) and a 5-level evaluation framework (component/pipeline/application/safety/operational) covering 15+ DeepEval metrics (Faithfulness, Answer Relevancy, Contextual Recall/Precision/Relevancy, Toxicity, PII Leakage, etc.)
- Implemented automated regression testing with metric-based gating (pass/fail/review) to detect quality and safety regressions across pipeline changes.

Deepfake Detection Web App | PyTorch, OpenCV, Streamlit (Live)

- Developed a Deepfake Image & Video Detection system using EfficientNet-B4 (CNN), trained on the 140K Real & Fake Faces dataset to classify images and videos as Real or Fake.
- Built and deployed a Streamlit web app for image input predictions, with model weights hosted on Hugging Face Hub and code on GitHub.
- Achieved 91%+ validation accuracy on the 140K dataset with EfficientNet-B4, evaluated using AUC-ROC and F1-score.

Sign Language Recognition System

- Built a CNN-based model to recognize hand gestures and convert them into text/speech; used OpenCV for real-time video processing. Technologies: Python, TensorFlow, OpenCV, CNN.
- Created a custom dataset of 5 gesture classes (Hello, Stop, Thank You, Rock, Perfect) and applied preprocessing (hand-region cropping, 64x64 resizing, normalization) and data augmentation to improve generalization.

EDUCATION

I.K. Gujral Punjab Technical University, Punjab

Bachelor of Technology in Computer Science and Engineering (AI & ML) Aug 2022 – June 2026

CERTIFICATIONS & LANGUAGES

Certifications: Python AI/ML, Generative AI Professional (Oracle)
Languages: Hindi (Native), English (Intermediate)